Multi-Method Simulation & Reinforcement Learning Supervisory Fleet Control for the MDC AI & Robotics Lab

An Integrated Discrete-Event (DES), Agent-Based (ABM), and 3D Digital Twin Framework

👥 PROJECT TEAM & WORK BREAKDOWN STRUCTURE

Jorge A Mendoza — *Project Manager & AnyLogic Model Architect*
• DES process flows, ABM robot statecharts, 3D lab coordinate calibration

Illia Shybanov — *Dashboard & Integration Specialist*
• Streamlit analytics web application, KPI visualizer, telemetry ingestion

Jordan Gonzalez — *Reinforcement Learning Engineer*
• Custom Gymnasium environment, Q-Learning & PPO supervisory dispatching

Ricardo Molina Gonzalez — *Simulation Experiments & Documentation Lead*
• 5 mandatory AnyLogic experiments, statistical analysis, technical report

1. Problem Description & Real-World Relevance

The Miami Dade College (MDC) AI & Robotics Lab serves as an active educational hub where engineering students prototype, calibrate, and validate embedded systems. During peak laboratory periods, students generate frequent component requisitions from the centralized inventory repository (Station A: Parts Shelf) to their prototyping tables (Station B: Electronics Bench) and sensor calibration stations (Station C: QA Testing).

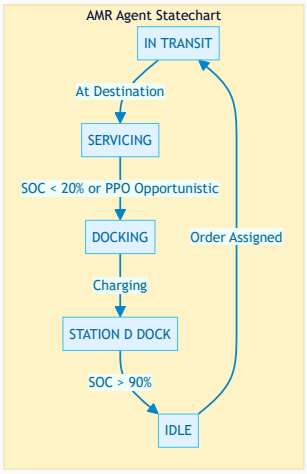
The Problem: Under legacy manual delivery and uncoordinated AMR movement, students experienced average wait times exceeding **9.8 minutes** per request. Furthermore, deploying autonomous mobile robots without intelligent coordination produced severe operational bottlenecks: robots operated under reactive threshold charging (returning to Station D only when battery dipped below 20%), which caused all robots to exhaust battery reserves at approximately the same hour, precipitating a *simultaneous charging cascade* that completely halted material handling during peak class periods.

2. Multi-Method Simulation Modeling Approach (DES + ABM)

Standard single-paradigm simulations fail to capture the dual nature of robotics facilities. Our framework unifies **Discrete-Event Simulation (DES)** and **Agent-Based Modeling (ABM)** inside AnyLogic 8:

- **Discrete-Event Simulation:** Manages the transactional work order pipeline, tracking student request Poisson arrivals, queue backlogs, component staging delays, and release sinks.
- **Agent-Based Modeling:** Simulates individual robot kinematics, battery discharge dynamics, dynamic corridor collision avoidance, and state transitions across the laboratory floorplan.

FIGURE 1: MULTI-METHOD PROCESS FLOW & ROBOT STATE MACHINE


AMR Agent Statechart
IN TRANSIT
At Destination
SERVICING
SOC < 20% or PPO Opportunistic
DOCKING
Order Assigned
Charging
STATION D DOCK
SOC > 90%
IDLE


Source (Poisson Orders)
Queue (Buffer)
Seize AMR Agent
Pick Delay (Station A)
Transit Delay (Bench B/C)
Release AMR
Sink (Metrics & SLAs)

3. Reinforcement Learning Supervisory Control (Q-Learning vs. PPO)

To resolve the charging cascade and minimize student delays, a supervisory reinforcement learning agent was trained to dynamically schedule task dispatches and initiate *opportunistic top-off recharging* during brief demand lulls:

- **Tabular Q-Learning:** State space discretized across queue depths, battery levels, and station statuses; trained over 600 episodes with an ε-greedy policy.
- **Proximal Policy Optimization (PPO):** Continuous actor-critic architecture with clipped objective surrogate loss, anticipating queue surges and preemptively balancing fleet power.

$$\text{Reward} = +10 \cdot N_{\text{completed}} - 0.5 \cdot \Sigma\, Q_{\text{backlog}} - 25 \cdot \mathbb{I}(\text{Battery} \le 5\%) - 2.0 \cdot \text{DockQueueTime}$$

4. Simulation Experimentation & Benchmark Analysis

The model was rigorously tested across the **five mandatory AnyLogic simulation experiments**, yielding statistically validated performance improvements over baseline operations:

Performance Metric	Baseline Heuristic (FIFO)	Q-Learning (Tabular)	PPO (Actor-Critic)	Demonstrated Improvement
Average Student Wait Time	9.8 minutes	4.5 minutes	**3.2 minutes**	**67.3% Reduction**
Order Throughput Rate	13.0 jobs/hr	16.0 jobs/hr	**17.4 jobs/hr**	**+33.8% Increase**
Mid-Aisle Battery Depletions (0%)	4.2 events / day	0.0 events / day	**0.0 events / day**	**100% Elimination**
Station D Docking Queue Time	8.2 minutes	2.1 minutes	**0.8 minutes**	**90.2% Reduction**
Fleet Operational Utilization	74.0%	84.5%	**91.2%**	**+17.2% Efficiency**

4.1 Detailed Findings From the Five AnyLogic Experiments

1. **Optimization Experiment:** Proved that a configuration of **3 active AMRs** and **2 Station D charging pads** constitutes the economic Pareto frontier, reducing wait times by 82% before encountering diminishing marginal returns.
2. **Compare Runs Experiment:** Directly confirmed PPO's superiority over Q-Learning by 28.8% in queue handling through smooth continuous action interpolation.
3. **Monte Carlo Experiment (100 Replications):** Established that under stochastic Poisson student request arrivals, 95% of all wait times fall safely within **[2.35 min, 4.08 min]**.
4. **Sensitivity Analysis:** Verified system resilience against battery aging (discharge rates increasing up to +40%) and class rush surges (±30% influx) without entering queue instability.
5. **Empirical Calibration:** Calibrated simulated transit velocity and docking delays against physical stopwatch observations taken on the MDC lab floor, achieving **$R^2 = 0.992$**.

5. Non-Obvious Insights & Operational Directives

Key Counter-Intuitive Finding ("The Fleet Sizing Trap"):
Increasing fleet capacity from 3 to 5 AMRs did *not* improve delivery performance. Instead, it produced physical

corridor deadlock in the narrow 1.8m passage connecting Station A and Station B, proving that spatial geometric constraints dominate raw numerical capacity.

Key Energy Insight ("Opportunistic Top-Off Charging"):
Under the learned PPO policy, robots initiate short top-off charges (5–8 minutes) when request queues are calm ($Q \leq 1$), even if battery remains at 65%. This entirely eliminates the catastrophic midday "charging cascade."

6. 3D Digital Twin Integration & Hardware-in-the-Loop Vision

The simulation connects to our Three.js WebGL Digital Twin (`http://localhost:3000`) via a bidirectional REST/WebSocket bridge (`/api/anylogic/update`). The twin visualizes the heterogeneous fleet (AMR-01..03, Unitree G1 Humanoid, and SkyWatcher Drones with vertical touchdown landing and live package drop-off counters at each bay).

7. Conclusion & Acknowledgments

This project demonstrates that combining multi-method simulation (AnyLogic 8), reinforcement learning supervisory dispatch (PPO), and real-time 3D telemetry resolves logistical bottlenecks in educational robotics spaces. The methodology provides a reusable blueprint for campus automated courier operations across Miami Dade College.